区间dp

人员

李瑞涵、田心一、柳力玮、刘祺、纪博涵、赵书梵 到课, 高健桓、初锦阳、刘宸熙、苑钊、杨咏丞 线上

上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/4033

王向东老师周日三点半C++dp练习

刷新

#	用户名	姓名	编程分	时间	A (12 / 15)	B (12 / 14)	C (12 / 18)	D (12 / 23)	E (11 / 23)	F (6 / 8)	G (3 / 3)
1	zhaoshufan	赵书梵	7	6:9:8	0:13:43	0:16:58	0:21:39	0:33:29	0:52:55 ♥	1:43:33 ♥	2:6:51 ♥
2	liuqi	刘祺	7	8:19:27	0:2:59 ♥	0:5:39 ♥	0:10:11 ♥	0:24:16 ♥	(-1) 0:56:27	(-1) 1:57:33	4:2:22
3	liuliwei	柳力玮	7	91:30:6	0:5:27	0:9:10	0:12:57	0:43:20	(-4) 29:25:52	29:56:18	29:37:2
4	tianxinyi	田心一	6	5:13:14	0:4:3	0:8:22	0:15:6	(-1) 0:43:24	1:22:35	(-1) 1:59:44	
5	gaojianhuan	高健桓	6	57:14:23	0:9:3	0:12:57	0:17:32	1:47:30	1:40:19	53:7:2	
6	yangyongcheng	杨咏丞	6	163:54:5	(-1) 0:13:52	(-1) 0:19:27	0:28:0	(-4) 5:4:41	54:41:30	101:6:35	
7	jiangshuzhang	蒋叔璋	5	3:32:28	0:9:54	0:13:3	(-1) 0:23:26	0:46:2	1:40:3		
8	liruihan	李瑞涵	5	4:11:49	0:5:21	0:8:47	0:15:6	(-2) 0:41:24	(-1) 2:1:11		
9	chujinyang	初锦阳	5	4:23:50	0:10:17	0:21:19	(-2) 0:43:18	0:57:52	1:31:4		
10	liuchenxi	刘宸熙	5	4:50:52	(-1) 0:6:34	0:11:51	(-1) 0:23:29	(-1) 0:40:13	(-1) 2:8:45		
11	yuanzhao	苑钊	5	5:14:19	(-1) 0:9:58	0:14:28	0:20:47	0:47:32	(-4) 2:1:34		
12	jibohan	纪博涵	4	4:53:21	0:7:3	(-1) 0:22:43	(-2) 1:4:16	(-3) 1:19:19	(-1)		

本周作业

https://cppoj.kids123code.com/contest/4165 (课上讲了 A ~ D 题, 课后作业是 E 题必做)

课堂表现

今天课上讲了区间 dp 的内容, 这个内容可能偏难一点, 同学们课上做题做的普遍不是很好, 课下尽量多复习一下。

课堂内容

[NOIP 2004 提高组] 合并果子

贪心, 每次取最小的两个合并

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
```

```
    priority_queue<int, vector<int>, greater<int>>q;
    int n; cin >> n;
    while (n -- ) { int x; cin >> x; q.push(x); }

    int res = 0;
    while (q.size() > 1) {
        int t1 = q.top(); q.pop();
        int t2 = q.top(); q.pop();
        int t = t1 + t2;
        q.push(t); res += t;
    }
    cout << res << endl;
    return 0;
}
```

石子合并（弱化版）

f[i][j]: 把 i~j 合并时, 最小花费多少

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 300 + 5;
const int inf = 0x3f3f3f3f;
int w[maxn], f[maxn][maxn];
int p[maxn];

int sum(int l, int r) { return p[r] - p[l-1]; }

int main()
{
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) cin >> w[i], p[i] = p[i-1] + w[i];

  for (int len = 2; len <= n; ++len) {
    for (int i = 1; i+len-1 <= n; ++i) {
      int j = i+len-1;
      f[i][j] = inf;
      for (int k = i; k < j; ++k)
        f[i][j] = min(f[i][j], f[i][k] + f[k+1][j] + sum(i,j));
    }
  }
  cout << f[1][n] << endl;
  return 0;
}
```

[NOI1995] 石子合并

整体思路跟上个题一致, 多一个破环成链的思想, 把数组扩两倍

```cpp
#include <bits/stdc++.h>

using namespace std;

const int maxn = 200 + 5;
const int inf = 0x3f3f3f3f;
int w[maxn], p[maxn];
int f1[maxn][maxn], f2[maxn][maxn];

int sum(int l, int r) { return p[r] - p[l-1]; }

int main()
{
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) cin >> w[i], w[n+i] = w[i];
  for (int i = 1; i <= 2*n; ++i) p[i] = p[i-1] + w[i];

  memset(f1, 0x3f, sizeof(f1));
  for (int i = 1; i <= 2*n; ++i) f1[i][i] = 0;

  for (int len = 2; len <= n; ++len) {
    for (int i = 1; i+len-1 <= 2*n; ++i) {
      int j = i+len-1;
      for (int k = i; k < j; ++k) {
        f1[i][j] = min(f1[i][j], f1[i][k] + f1[k+1][j] + sum(i,j));
        f2[i][j] = max(f2[i][j], f2[i][k] + f2[k+1][j] + sum(i,j));
      }
    }
  }

  int minn = 1e9, maxx = 0;
  for (int i = 1; i <= n; ++i) {
    minn = min(minn, f1[i][i+n-1]);
    maxx = max(maxx, f2[i][i+n-1]);
  }
  cout << minn << endl;
  cout << maxx << endl;
  return 0;
}
```

[IOI 2000] 回文字串

f[i][j]: 把 i~j 变为回文时, 最少需要插入几个字符

当 s[i]==s[j] 时, f[i][j] = f[i+1][j-1]

当 s[i]!=s[j] 时, f[i][j] = min(f[i+1][j], f[i][j-1]) + 1

```cpp
#include <bits/stdc++.h>
```

```
using namespace std;

const int maxn = 1000 + 5;
char s[maxn];
int f[maxn][maxn];

int main()
{
    cin >> (s+1);
    int n = strlen(s+1);
    for (int len = 2; len <= n; ++len) {
        for (int i = 1; i+len-1 <= n; ++i) {
            int j = i+len-1;
            if (s[i] == s[j]) f[i][j] = f[i+1][j-1];
            else f[i][j] = min(f[i+1][j], f[i][j-1]) + 1;
        }
    }
    cout << f[1][n] << endl;
    return 0;
}
```

[USACO06FEB] Treats for the Cows G/S

f[i][j]: 最后卖 i~j 这一段时, 最多能得到多少钱

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 2000 + 5;
int w[maxn], f[maxn][maxn];

int main()
{
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) cin >> w[i], f[i][i] = w[i]*n;

  for (int len = 2; len <= n; ++len) {
    for (int i = 1; i+len-1 <= n; ++i) {
      int j = i+len-1, day = n-len+1;
      f[i][j] = max(f[i][j-1]+w[j]*day, f[i+1][j]+w[i]*day);
    }
  }

  cout << f[1][n] << endl;
  return 0;
}
```